Public Capital Investment Frictions: A Piece in the Euler Equation Puzzle

Yichu Li*

Link to Latest version

Last updated April 17, 2026

Abstract

The text of your abstract.

Keywords: Growth, Euler Equation Puzzle, Public Investment, Development Accounting

*The London School of Economics and Political Science. y.li395@lse.ac.uk.

1 Introduction

Much of development economics research attempt to understand what is holding back economic development in lower- and middle-income countries. The categorisation of countries into 'developing' versus 'developed' implies that the former have higher development potentials. The Solow (1956) growth model predicts exactly this pattern because poorer economies with smaller capital stocks should face higher marginal returns to capital investment and therefore grow faster than richer economies. While the broad pattern of higher GDP growth rates in lower-income countries over the past four decades is consistent with this prediction, many countries have still believed to been growing at rates below their potentials.

Lucas (1990) and Caselli and Feyrer (2007) points out that the higher marginal product of capital in lower-income countries is often balanced-off by their lower endowment in complimentary factors like human capital, or lower total factor productivity. This provides an aggregate-level explanation for why capitals do not flow from higher-income to lower-income countries as much as expected. However, the puzzle is not fully addressed because on the aggregate-level, it does not explain why lower-income countries do not invest in human capital or R&D to benefit from the high capital returns. Moreover, through micro-level RCTs, de Mel et al. (2008) show that microenterprises in Sri Lanka face 4.6%-5.3% real returns to capital, and Balboni et al. (2022) show significant long-run returns on agricultural capitals sufficient to push households in Bangladesh out of a poverty trap. Kremer et al. (2019) uses an household-level intertemporal constrained optimisation model to illustrate an '*Euler Equation Puzzle*' where people in lower-income settings invest sub-optimally given high returns from capital investment.

Relating to this paradox, a large body of literature including Balboni et al. (2022) and research covered in Banerjee (2003) point to liquidity and credit constraints facing households and firms; some including Acemoglu et al. (2001), Mauro (1995) and Banerjee et al. (2002) point to institutional ineffectiveness and insufficient property rights protections; while Kremer et al. (2019) point to studies in behavioural channels such as present bias. However, the existing literature focus primarily on private investment channels. For example, these are scenarios where farmers investing in a cattle or fertilisers and benefit fully from the output gains. Meanwhile, returns on certain capitals like infrastructure or human capital are also high in lower-income countries. Investment in public capitals often require fiscal spending from the government funded primarily by tax revenues. Besley and Persson (2013) show that in lower-income countries taxation is more difficult administratively while frictions against efficient spending are stronger.

Nonetheless, it might still seem irrational for above-subsistence households in lower-income countries to avoid taxes even with low expected penalties because the long-term returns from public capitals can be sufficiently high. This paper shows that the set-up of the taxation-to-government-spending mechanism can produce a collective action problem at the household level when perceived state taxation capacity is low.

In Section 2 of this paper, I will illustrate the collective action problem on public capital investment base on the intertemporal constrained optimisation model of Kremer et al. (2019).

2 A Household Intertemporal Constrained Optimisation Model Incorporating Public Capitals

I build on the intertemporal optimisation framework of Kremer et al. (2019). I first recall the model setup and present the Euler Equation Puzzle through the model. I then consider the extreme scenario where only public capitals exist and investment occurs only through taxation, showing that this generates a collective action problem and a government efficiency problem that dilutes the effective return to capital when state taxation capacity and efficiency is low. Finally, I will extend this analysis into a generalised model incorporating both private and public capital, allowing households to allocate investments between the two.

2.1 Baseline: Standard Euler Equation

Consider in a deterministic discrete-time intertemporal consumption model, households maximise their discounted lifetime utility over an infinite time horizon.

Optimisation Problem

Households face the following optimisation problem:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \delta^t u(C_t), \tag{1}$$

where $0 < \delta < 1$ is the annual discount factor, C_t is the household consumption in period t, and $u(C_t)$ measures the utility from consuming C_t in period t. $u(C)$ is increasing and concave and follows Inada Conditions: $u'(C) > 0$, $u''(C) < 0$, $\lim_{C\to 0} u'(C) = \infty$, and $\lim_{C\to\infty} u'(C) = 0$.

Budget Constraint

$$x_{t+1} = f'(K_t)(x_t - C_t) \tag{2}$$

x_t is household assets ('cash on hand') in period t with x_0 given, C_t is consumption in period t, and $f'(K_t)$ is the gross return to capital. $f(K)$ is increasing and concave, following: $f'(K) > 0$, $f''(K) < 0$, $\lim_{K\to 0} f'(K) = \infty$, and $\lim_{K\to\infty} f'(K) = 1$ (long-run steady state).

Discrete Time Euler Equation

Solve for optimal intertemporal consumption path:

$$u'(C_t) = \delta f'(K_t) u'(C_{t+1}), \tag{3}$$

and assume constant-elasticity-of-substitution (CES) utility:

$$u(C) = \begin{cases} \frac{C^{1-\sigma}}{1-\sigma} & \text{if } \sigma \neq 1 \\ \ln(C) & \text{if } \sigma = 1 \end{cases}$$

so that the elasticity of intertemporal substitution is constant and independent of C_t. This gives the consumption growth rate:

$$\frac{C_{t+1}}{C_t} - 1 = (\delta f'(K_t))^{\frac{1}{\sigma}} - 1, \tag{4}$$

where $\frac{1}{\sigma}$ is the elasticity of intertemporal substitution.

The Puzzle

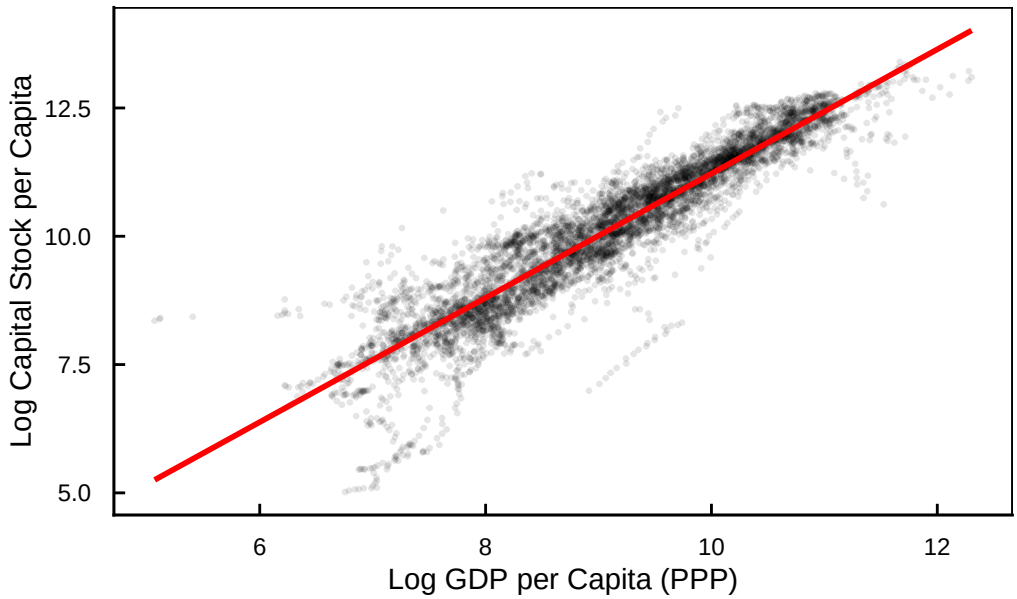
Assume that the model governs the intertemporal consumption behaviour of a representative household, the model should also apply at the macro level. To illustrate the puzzle with macro-level data, I use $\delta = 0.96$ [1]and $\sigma = 2$ [2] proposed in Kremer et al. (2019), and internal rates of return (IRR) from PWT [3] (Feenstra et al., 2015) as a proxy for $f'(K)$[4] to model the predicted consumption growth rates under Equation 4. Intuitively, a concave production function means that countries with lower capital stock face higher returns on capital investment. Figure 1 shows that countries with lower income have lower capital stock. Therefore, according to Equation 4, consumption growth rates should be high. However, Figure 2 Panel (A) shows that in many countries, the actual growth rates is insensitive to IRR and many fall below the model predictions. Moreover, Figure 2 Panel (B) shows a perplex trend where countries with higher predicted consumption growth rates tend to have larger gap between predicted and actual rates. This pattern is most pronounced in countries such as Egypt, Venezuela, Côte d'Ivoire, and Zimbabwe. Therefore, the core of the puzzle are why households in these economies invest at levels far below what the standard model predicts as optimal, and why is this pattern more pronounced in countries where returns on capitals are high.

2.2 Capital as a Public Good

The baseline model in Section 2.1 assumes capital investments are private for households. This means that households receive the full returns $f'(K)$ because they would own the invested capital in productions. However, capitals with public good features like infrastructure often instead require public investment commonly through government spending, and the benefits from these investments are shared among the population. Aschauer (1989) estimates a high average elasticity of 0.39 between output and public investment in the United States over 1949-1985, while Bom and Ligthart (2014) proposes a more modest estimate of 0.106. The latter estimate that the short-

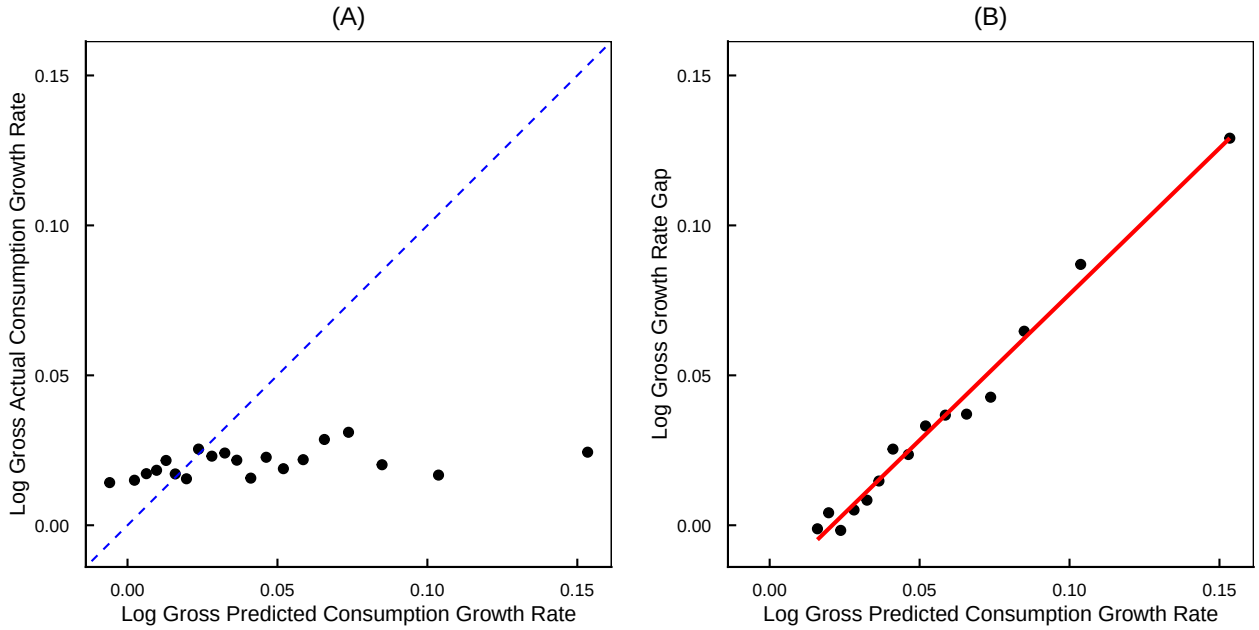
[1] $\frac{1}{\delta} - 1 \approx 0.04$, implying a 4% annual rate of time preference.
[2] Elasticity of intertemporal substitution of 0.5.
[3] Available for download at www.ggdc.net/pwt
[4] See https://www.rug.nl/ggdc/docs/the_next_generation_of_the_penn_world_table.pdf

Figure 1: Lower Capital Stock in Lower-Income Countries



Notes: Red line is OLS fit. Panel data from 1950 to 2023. Capital stock in millions US$ at 2021 PPPs. Output−side real GDP in millions US$ at 2021 chained PPPs. Source: PWT 11.0.

Figure 2: Actual vs. Predicted Consumption Growth

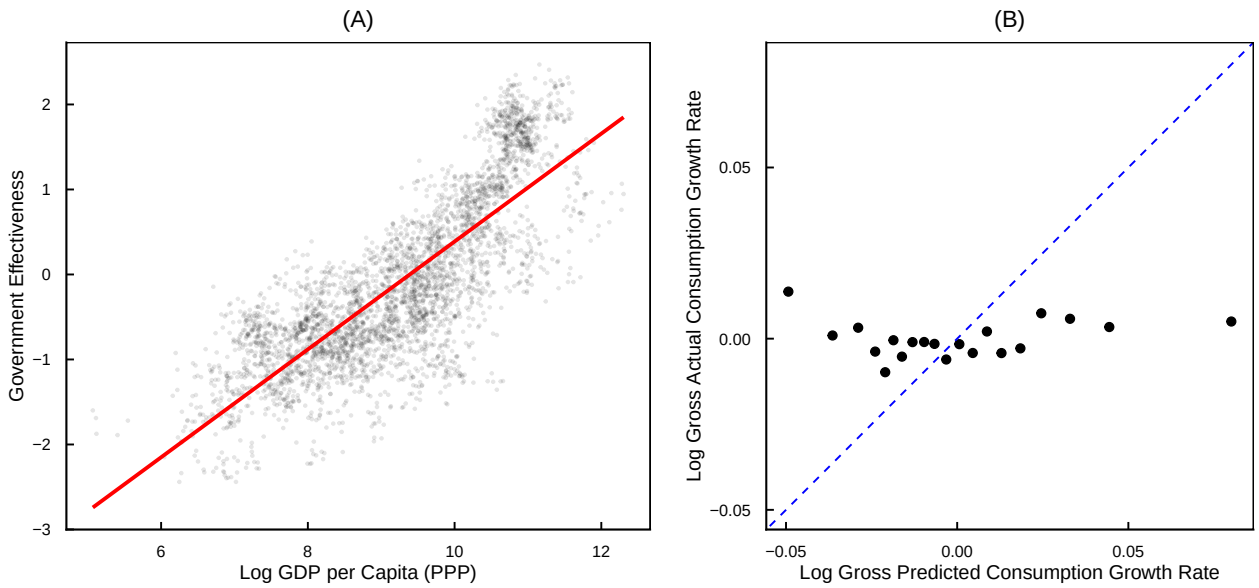


Notes: Binned scatter plots with 20 equal−frequency bins. The blue dashed line is the 45 degree line. The red line is OLS fit. Countries with populations below 1 million and irregular internal rates of return excluded. Data after 2019 is excluded to avoid Covid−19 distortions. Source: PWT 11.0.

run national-level elasticity of 0.083 still imply a 16% marginal return to public capital in OECD countries. The returns are likely higher in regional levels and much higher in lower-income countries with lower public capital stocks. However, Besley and Persson (2013) show likely political economy reasons including political instability, favourtism, inequality, or corruptions which undermine the taxation-to-government-spending mechanism necessarily for optimal public capital investment. These factors are more prominent in lower-income countries, while countries might improve taxation capacities and have reduced spending frictions over development.

However, if public capital investments yield high returns, why would rational households even not pay taxes voluntarily but choose to avoid them when possible? Indeed, findings in Mauro (1995) also show low public (and private) investment levels when governments are inefficient, and Figure 3 Panel (A) shows that government effectiveness is lower in lower-income countries. Therefore, this high government spending friction could discourage households to pay taxes if they believe the revenue is not put in use efficiently. Nonetheless, Figure 3 Panel (B) shows that the trend in Figure 2 Panel (A) persist even controlling for government effectiveness. My following model will present that the household level, the fiscal features of lower-income countries can also create a collective action problem so that they optimally avoid paying taxes individually, thereby producing a global sub-optimal outcome.

Figure 3: Gap Persists after Controlling for Government Effectiveness



Notes: Red line is OLS fit. Panel data from 1996 to 2023. Output−side real GDP in millions US$ at 2021 chained PPPs. Panel B shows binned scatter plots with 20 equal−frequency bins after partialling out government effectiveness. The blue dashed line is the 45 degree line. Source: PWT 11.0, WGI. Source: WDI.

To illustrate, I will present an extreme case where we assume capitals are all publicly invested through taxation by governments. Consider the same optimisation problem as Equation 1 with a different budget constraint for homogeneous households in an economy. For any household i, the

budget constraint is:

$$x_{i,t+1} = \frac{1}{N} \cdot f'(K_t) \cdot [(x_{i,t} - C_{i,t}) + \sum_{\forall j \neq i} (x_{j,t} - C_{j,t})], \tag{5}$$

where N is the total number of households in the economy, and $\sum_{\forall j \neq i}(x_{j,t} - C_{j,t})$ is the aggregate public investment from all other households. Because public capitals can only be invested collectively, the total returns depend on the aggregate investment level. For simplicity, assume each individual household receive an equal share of the total returns, public capital investment is the only government expenditure, and the government has perfect efficiency.

Weak Taxation Capacity

First consider a scenario where the state taxation capacity is weak. To reflect weak taxation capacity, assume that household i makes public investment decision through taxation voluntarily and independent of other households' decisions. From the perspective of household i, the total returns from public capital investment as a public good is diluted among the population. Alternatively, for a fixed level of total returns on public capital, the impact of the household's investment level $(x_{i,t} - C_{i,t})$ on its assets in the next period is small for large N.

Solving for optimal intertemporal consumption path for household i given $\sum_{\forall j \neq i}(x_{jt} - C_{jt})$:

$$u'(C_{i,t}) = \frac{1}{N} \cdot \delta f'(K_t) u'(C_{i,t+1}). \tag{6}$$

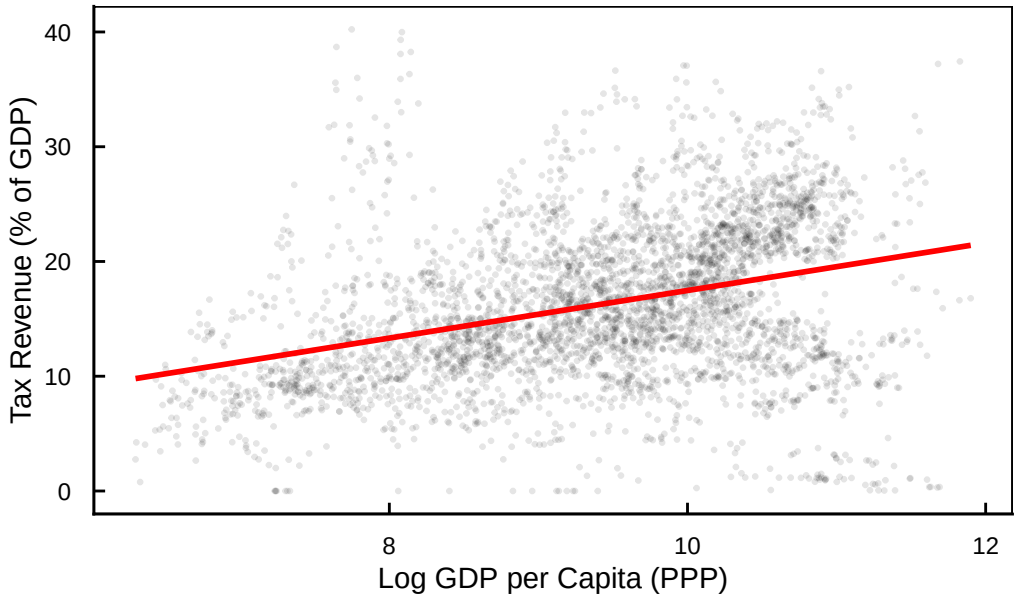
Notice that when $N = 1$, this equation is identical to Equation 3, but $u'(C_{i.t})$ decreases as N becomes larger. For increasing and concave utility function, $u'(C_{i,t})$ is smaller when $C_{i,t}$ is larger. This means that household i will consume more and invest less as N increases. For sufficiently large N so that the Euler Equation cannot be satisfied by an interior solution, household i will optimally choose a corner solution of consuming all and invest none. $\frac{1}{N}$ produces a collective action problem where the global optimal decision is for all households to invest and benefit from high $f'(K)$ but individuals households have a dominant strategy of not investing.

Figure 4 shows the countries with lower GDP per capita have lower tax revenue relative to their GDP. Therefore, this scenario is more consistent with lower-income settings and it is consistent with the reluctant tax compliance attitude among many low- and middle-income settings. This framework also provides an explanation to the prima facie irrationality of tax avoidance in lower-income countries where people can benefit largely from public investments.

Strong Taxation Capacity

Higher income countries tend to have stronger taxation capacity because, for example, income tax can be more easily collected from employees (Jensen, 2022). Although, public investment is less voluntary for tax rates as given, households might make public investment decisions by influencing

Figure 4: Higher Tax Revene in Higher-Income Countries



Notes: Red line is OLS fit. Panel data from 1972 to 2023. Output−side real GDP in millions US$ at 2021 chained PPPs. Source: WDI.

the tax rates through political channels. The key difference is that they are more likely to expect other households to also comply with the decided upon tax rates.

Consider a simple case of homogeneous households facing the same optimisation problem as Equation 1 but assume instead perfect enforcability upon any chosen taxation level and perfect government efficiency. Any individual household i chooses a taxation level (public investment level) $(x_{i,t} - C_{it})$ and expect symmetry from other households. Because households are homogeneous, the following should hold:

$$[(x_{i,t} - C_{i,t}) + \sum_{\forall j \neq i} (x_{j,t} - C_{j,t})] = N \cdot (x_{i,t} - C_{i,t}),$$

so that the budget constraint becomes identical to Equation 2:

$$x_{i,t+1} = \frac{1}{N} \cdot f'(K_t) \cdot N \cdot (x_{i,t} - C_{i,t}) = f'(K_t)(x_{i,t} - C_{i,t}).$$

Solving for optimal consumption path will produce an Euler Equation identical to Equation 3. Notice that the collective action problem is resolved in this setting because the perfect tax enforcement enables households to optimise consumption conditional on the expectation of symmetrical behaviours from others.

2.3 A Generalised Model

The collective action problem explored above is an example channel to explain why public capital investment frictions exist. In reality, there are likely many interacted channels and the specific functional form is difficult to define. Even for the collective action model, it is very reasonable to believe that the severity of the problem is not sensitive to the population size in a country linearly. Nonetheless, the public capital investment frictions at the household-decision level can be generalised as a wedge $\phi_{pub} \in [0, 1]$ where lower values indicate larger frictions. Moreover, recall that the literature also highlights various channels for the private-side frictions, so that the private-side frictions can be also generalised by $\phi_{priv} \in [0, 1]$ analogously. A generalised intertemporal household opitmisation model with both private and public capital returns has the following budget constraint:

$$x_{t+1} = r_t^{eff} \cdot (x_t - C_t), \tag{7}$$

where

$$r_t^{eff} = (1 - \tau) \cdot \phi^{priv} \cdot f'(K_t^{priv}) + \tau \cdot \phi^{pub} \cdot f'(K_t^{pub}),$$

with K^{priv} denoting private capital stock and τ denoting the share of the total investment amount that the household chooses to invest in public capital. Notice that households choose τ freely to maximise their budget constraint. Because $f'(K_t^{priv})$ and $f'(K_t^{pub})$ are concave, this means that at equilibrium:

$$\phi^{priv} \cdot f'(K_t^{priv}) = \phi^{pub} \cdot f'(K_t^{pub})$$

and rearranging:

$$\frac{\phi^{pub}}{\phi^{priv}} = \frac{f'(K_t^{priv})}{f'(K_t^{pub})} \tag{8}$$

shows that the ratio between the public and private capital investment frictions can be given by the ratio between the marginal investment returns from private and public capital.

Solving for optimal consumption path under the above constraint produces an Euler Equation:

$$u'(C_t) = \delta \cdot r^{eff} \cdot u'(C_{t+1}),$$

and the consumption growth rate is given by:

$$\frac{C_{t+1}}{C_t} - 1 = (\delta \cdot r^{eff})^{\frac{1}{\sigma}} - 1. \tag{9}$$

Recall the pattern illustrated in Figure 2 Panel (B): countries with higher predicted consumption growth rates tend to exhibit larger gaps between predicted and actual growth. Under the above model, this can be expected even if the private investment channel is frictionless.

3 Accounting the Public Capital Investment Frictions

The theoretical model features four structural parameters:

$$\Theta = \{\delta, \sigma, \tau, \phi\},$$

where δ is the annual discount factor, σ is the risk aversion coefficient and $\frac{1}{\sigma}$ is the elasticity of intertemporal substitution, τ is the share of total investment in public capital, and ϕ is the severity of the collective action problem for public capital investment. To discipline the model, I exogenously fix $\delta = 0.96$ and $\sigma = 2$ proposed by Kremer et al. (2019) based on the literature, same with the exercise in Figure 2. Although in the theoretical model, τ is chosen by the households, the public capital investment shares from ICSD can be interpretated as the equilibrium outcome of this choice.

To use macroeconomic data, I proxy for the returns on private and public capitals households face by the following Cobb-Douglas production function facing a representative household in the economy:

$$Y_{it} = A_{it}(K_{it}^{priv})^{\alpha^{priv}}(K_{it}^{pub})^{\alpha^{pub}}L_{it}^{1-\alpha^{priv}-\alpha^{pub}},$$

where the first-order conditions are:

$$f'(K_{it}^{priv}) = \alpha_{priv} \cdot \frac{Y_{it}}{K_{it}^{priv}} \quad \text{and} \quad f'(K_{it}^{pub}) = \alpha_{pub} \cdot \frac{Y_{it}}{K_{it}^{pub}}.$$

Taking the conservative estimate of short-run national-level output elasticity of public capital 0.083 gives:

$$f'(K_{it}^{pub}) = 0.083\left(\frac{Y_{it}}{K_{it}^{pub}}\right)$$

Assume for simplicity the labour market is competitive in aggregate,

$$w = \frac{\partial Y}{\partial L} = (0.917 - \alpha^{priv})\frac{Y}{L},$$

so that:

$$\alpha^{priv} = 0.917 - \frac{wL}{Y},$$

where the PWT provides the data on the labour share of income. This approach would give a mean output elasticity of labour around 0.523, and a mean ouput elasticity of private capital around 0.394.

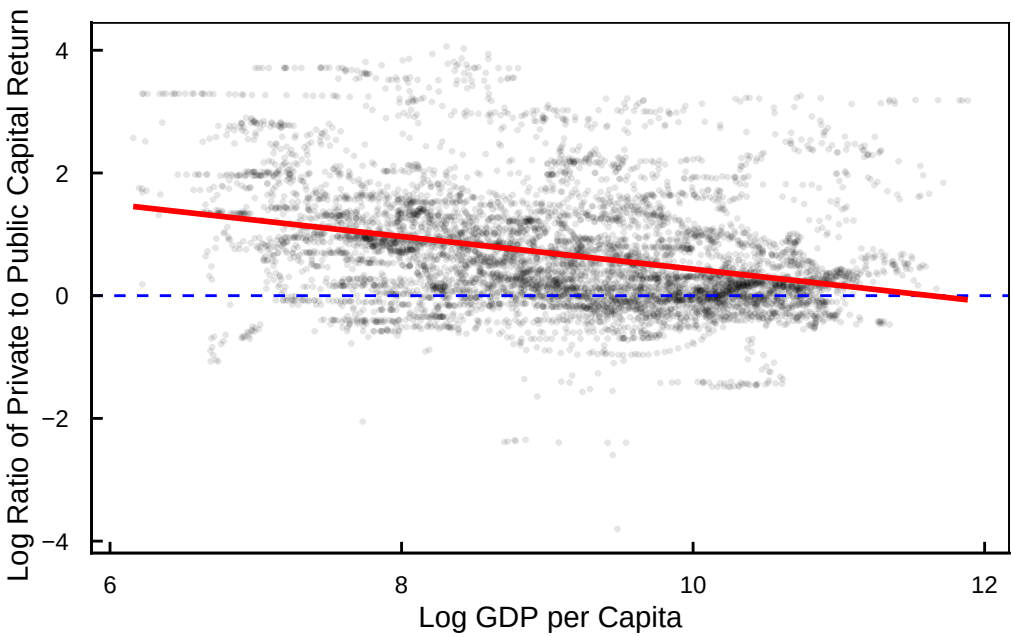
3.1 Ratios between Private and Public Capital Returns

Recall the equilibrium condition of Equation 8, I can rewrite:

$$\frac{\phi_{it}^{pub}}{\phi_{it}^{priv}} = \frac{f'(K_{it}^{priv})}{f'(K_{it}^{pub})} = \frac{\alpha^{priv}}{\alpha_{it}^{pub}}\left(\frac{K_{it}^{pub}}{K_{it}^{priv}}\right)$$

Using the above condition and the conservative output elasticity of public capital 0.083, Figure 5 shows most countries across time have larger public-side wedge compared to the private side, where the marginal returns on private capital investment is higher than public capital investment. This is an indication that the private capital investment frictions are higher on average across levels of development. (**INSERT SENSITIVITY TEST)** However, the red OLS fit shows that this pattern is more prominent on the lower log per capital GDP end, and it attenuates towards the higher end. This pattern is also robust after controlling for country and year fixed effects. **(INSERT ANNEX)** Recall Figure 2 that lower-income countries tend to have higher overall frictions, Figure 5 may suggest that the public capital investment frictions contribute less to the overall high frictions in lower-income countries, and the private-side frictions are more likely to be the driving force.

Figure 5: Ratio of Private to Public Capital Returns by Log GDP per Capita



Notes: Dashed line is when the ratio is 1 (log = 0). Red line is OLS fit. Sources: PWT 11.0, ICSD.

3.2 Incidental Motivating Evidence from Self-Employment Share

Ideally, ϕ^{pub} may be approximated by a variable that is differentially associated with the public capital investment channel but not with the private channel, and Section 2.2 and Besley and Persson (2013) show that it should likely relate to government taxation capacities. The self-employment share in total employment might be a proxy to consider in this context. Jensen (2022) identifies that the income tax exemption threshold co-moves with the employee/self-employed composition over development, such that the tax base consistently covers segments where third-party enforceable employee income predominates (85–95% employee share). This is primarily because that taxing self-employment is administratively more difficult, so governments target only employee incomes to protect tax morale. This pattern is consistent with Table 1 Panel (A) which shows higher self-employment percentage in total employment significantly associates with lower log public capital

Table 1: Self-Employment and Capital Investment

	Pooled	Country FE	Year FE	Country + Year FE
Panel A: $\log I_{it}^{pub}$				
Self-Employment (%)	-0.044***	-0.069***	-0.043***	-0.027*
	(0.002)	(0.014)	(0.002)	(0.013)
Sample Size	4674	4674	4674	4674
Within R^2		0.129	0.615	0.018
Panel B: $\log I_{it}^{priv}$				
Self-Employment (%)	-0.051***	-0.073***	-0.051***	-0.026*
	(0.002)	(0.014)	(0.003)	(0.011)
Sample Size	4674	4674	4674	4674
Within R^2		0.159	0.640	0.021

Notes: Dependent variable is log per capita investment. Standard errors clustered at the country level in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

investment across specifications. However, Table 1 Panel (B) shows that the pattern is very similar for the private capital investment channel. This means either self-employment share is correlated with both the private and public capital investment frictions through different mechanisms, or the frictions from both sides are highly correlated. Although this means that self-employment share is likely a bad proxy for ϕ^{pub} specifically, the similar coefficients across both channels suggest that self-employment share provides a symmetric signal to both channels. This motivates a test on the correlation between both friction channels and a grid search exercise utilising the symmetric signal from self-employment share.

3.3 Limited Cross-Channel Frictions Spillovers

Countries with high self-employment shares tend to be lower-income countries that share characteristics like prevalent liquidity constraints, property rights insecurity, or institutional ineffectiveness that could affect both the private and public capital investment frictions simultaneously. Table 2 shows Granger tests on first-differenced [5] log marginal returns from private and private capital. Changes in public capital marginal returns exhibit significant positive autocorrelation (lagged own coefficient 0.17–0.23), while the private side does not. This asymmetry suggests slower adjustment in public investment, meaning that public capital investment frictions are sticker, while the market adjusts faster on the private side. This is consistent with factors like political inertia, long infrastructure project duration, discontinuous government budgets which do not impact private-side investments. There are statistically significant small spillovers from the public capital investment frictions to the private side, but the reverse are even smaller in magnitude and less robust across specifications. This suggests although spillovers between the private and public friction channels

[5] Im-Pesaran-Shin panel tests fail to reject the null of a unit root for both $\log f'(K^{priv})$ ($p = 0.28$) and $\log f'(K^{pub})$ ($p \approx 1.00$) in levels, but strongly reject in first differences ($p < 0.001$ for both).

Table 2: Granger Tests

	Pooled	Country FE	Year FE	Country + Year FE
Panel A: $\Delta log f'(K_{it}^{priv})$				
$\Delta log f'(K_{i,t-1}^{pub})$	0.093***	0.104***	0.086***	0.097***
	(0.023)	(0.024)	(0.023)	(0.024)
$\Delta log f'(K_{i,t-1}^{priv})$	0.014	-0.015	0.006	-0.023
	(0.018)	(0.018)	(0.018)	(0.018)
Sample Size	6417	6417	6417	6417
Within R^2		0.005	0.005	0.004
Panel B: $\Delta log f'(K_{it}^{pub})$				
$\Delta log f'(K_{i,t-1}^{pub})$	0.228***	0.186***	0.214***	0.167***
	(0.018)	(0.018)	(0.018)	(0.018)
$\Delta log f'(K_{i,t-1}^{priv})$	-0.037**	-0.022	-0.044**	-0.027*
	(0.014)	(0.014)	(0.014)	(0.014)
Sample Size	6417	6417	6417	6417
Within R^2		0.028	0.032	0.020

Notes: First difference removes AR(1) structure. Standard errors clustered at the country level in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

exist, the frictions also operate largely via different mechanisms. Therefore, this is also motivating evidence to study the public capital investment channel separately from the private channel.

3.4 Grid Search Exercise

Since Table 1 shows symmetric associations between self-employment share and log private and public investment levels, I use this signal from self-employment share to explore the relative importance of either channel in explaining the consumption gap. Using self-employment share as a proxy for ϕ^{priv} and ϕ^{pub} and to allow different values across countries and time, I parameterise both wedges for country i in year t with the following logistic function to be bounded between 0 and 1:

$$\phi_{it} = \frac{1}{1 + \exp(\beta_0 + \beta_1 \cdot SelfEmp_{it})} \tag{10}$$

I assume perfect foresight so that the Euler equation yields $C_{i,t+1}/C_{it} = (\delta \cdot r_{i,t+1}^{eff})^{1/\sigma}$. For estimation, I take logs on both sides to render the estimating equation additive and stabilise the variance of consumption growth:

$$\ln(\frac{C_{i,t+1}}{C_{i,t}}) = \frac{1}{\sigma} \ln(\delta \cdot r_{it}^{eff}(\beta_0, \beta_1)) + \varepsilon_{it},$$

where

$$r_{it}^{eff} = 1 - d_{it} + (1 - \tau_{it}) \cdot \phi_{it}^{priv} \cdot f'(K_{it}^{priv}) + \tau_{it} \cdot \phi_{it}^{pub} \cdot f'(K_{it}^{pub})$$

is the gross effective return per unit of savings, combining undepreciated capital and marginal products net of channel-specific frictions. The residual term ε_{it} captures all other possible channels such as credit constraints or institutional features and d_{it} denotes capital depreciation rates from the PWT data.

I perform two grid search exercises assuming one channel to be frictionless and use Equation 10 to proxy for the wedge of the other channel. The grid search will find a combination of β_0 and β_1 to minimise the sum of squared error (SSE) of the difference between log consumption growth and the model predictions under the specified range. Under the same grid search space, Figure 6 Panel (A) shows that assuming frictionless private investment channel, the grid search cannot find a unique interior combination of the coefficients to reduce the gap between actual consumption growth and model predictions effectively via only the public channel **(INSERT ANNEX)**; Panel (B) assuming instead that the public channel is frictionless, the grid search finds an optimal combination of $\beta_0 = 0.2$ and $\beta_1 = 0.015$. The heat map in Panel (B) shows a clear convergence pattern to an interior solution, while Panel (B) shows substantially flatter SSE surface. There are four following possible explanations for this asymetric pattern.

1. The functional form is significantly mi-specified;
2. The signals from self-employment share are weak in relation to public capital investment frictions;
3. Cross-country and -time variations on public capital investment frictions are too weak;
4. The public capital investment frictions channel is weaker in explaining the overall frictions.

Responding to the first explanation. The flatter SSE surface in Figure 6 Panel (A) than Panel (B) indicates that plugging the self-employment proxy in the private friction channel absorbs the error term better than in the public friction channel. However, if there are omitted factors that associate with self-employment but affect one channel more than the other, the functional form in this exercise may absorb these omitted effects asymmetrically when effects are restricted to act via only one channel each time. In that case, the convergence in Panel (B) could partly reflect self-employment capturing omitted channels rather than the private-side frictions genuinely explaining more of the overall gap. Nonetheless, Table 1 suggests that the signal from self-employment affects the private and public investments similarly which could indicate similar associations between omitted factors and both private and public friction channels. The only caveat is that factors signaled by self-employment share might associate with different forms of productivity which could affect returns from private or public capital returns differently, and Table 1 cannot rule out this mechanism. Moreover, the asymmetric pattern cannot be attributed to the logistic proxy functional form because it should have impacted both channels rather than only the public channel.

The second and third explanation can be plausibly ruled out. Table 1 shows symmetric significant associations between self-employment share and both private and public investment

channels, and Figure 6 Panel (B) shows that the grid search range provides sufficiently well-defined SSE surface for convergence. Therefore, it is unlikely that self-employment provides sufficiently strong signal for the private channel but not the public channel. Moreover, the within R^2 rows in Table 1 also shows sufficient cross-country and -time variations under year and country fixed effect specifications.

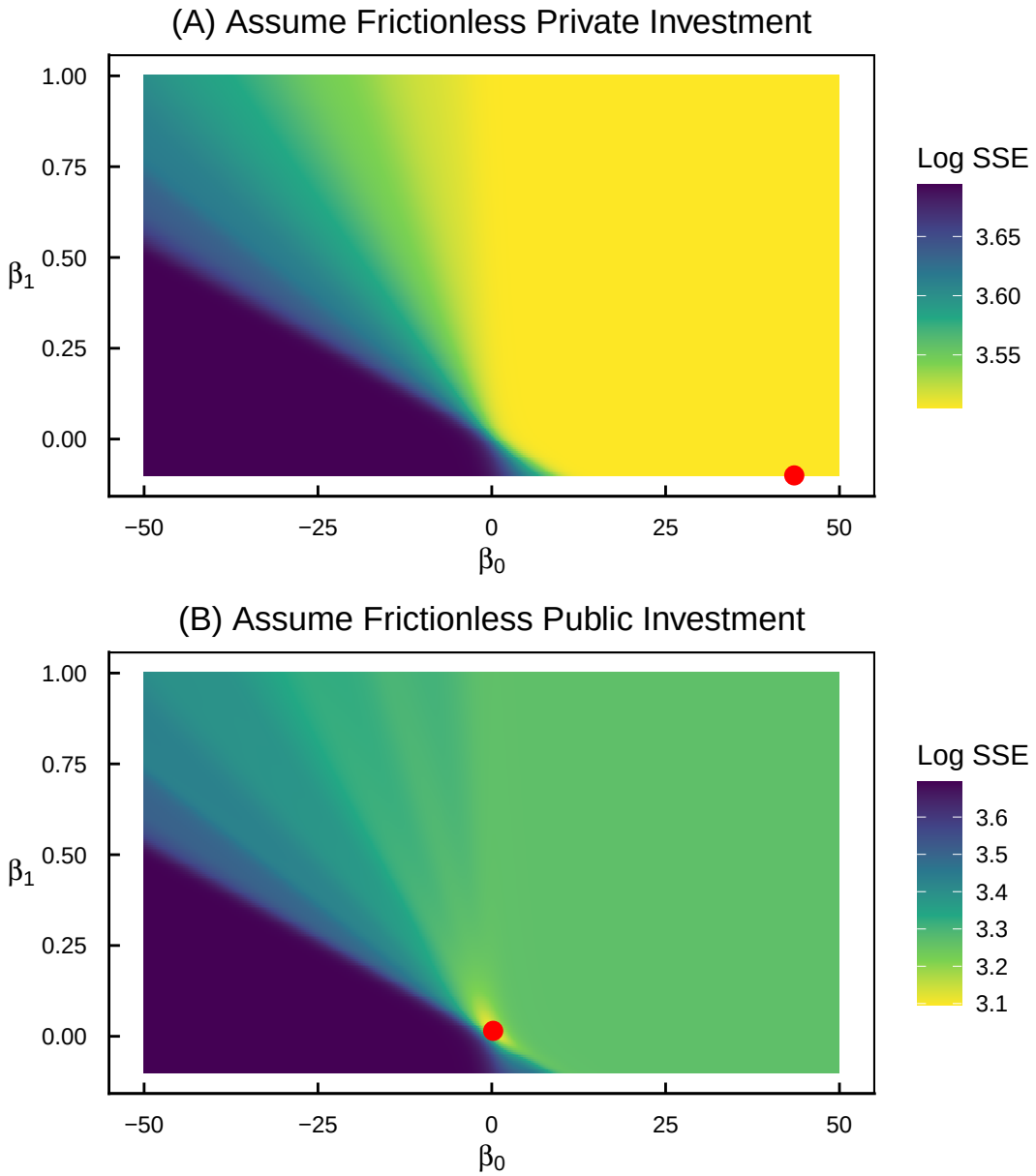
Therefore, the pattern from the grid search exercises most plausibly reflect the fourth explanation that the private capital investment friction channel explains more of the consumption growth rate gap than the public friction channel. This is also consistent with the pattern in Figure 5.

3.5 Bounds of Private vs. Public Capital Investment Frictions

To further the findings in Section 3.1 and Section 3.4, I perform a bounding exercise with aggregate data to quantify to what extent the private and public friction channel can possibly explain the consumption growth gap.

4 Implications

Figure 6: Grid Search



Notes: Heatmap of log sum of squared residuals. Panel (A) places the wedge on the public capital channel, Panel (B) on the private channel. Red dot indicates the parameter combination minimising SSE. Sources: PWT 11.0, ICSD, WDI.

References

Acemoglu, Daron, Simon Johnson, and James A Robinson (2001) "The Colonial Origins of Comparative Development: An Empirical Investigation," *American Economic Review*, 91 (5), 1369–1401, 10.1257/aer.91.5.1369.

Aschauer, David Alan (1989) "Is Public Expenditure Productive?" *Journal of Monetary Economics*, 23 (2), 177–200, 10.1016/0304-3932(89)90047-0.

Balboni, Clare, Oriana Bandiera, Robin Burgess, Maitreesh Ghatak, and Anton Heil (2022) "Why Do People Stay Poor?" *The Quarterly Journal of Economics*, 137 (2), 785–844, 10.1093/qje/qjab045.

Banerjee, Abhijit V (2003) "Contracting Constraints, Credit Markets, and Economic Development."

Banerjee, Abhijit V., Paul J. Gertler, and Maitreesh Ghatak (2002) "Empowerment and Efficiency: Tenancy Reform in West Bengal," *Journal of Political Economy*, 110 (2), 239–280, 10.1086/338744.

Besley, Timothy and Torsten Persson (2013) "Taxation and Development," in *Handbook of Public Economics*, 5, 51–110: Elsevier, 10.1016/B978-0-444-53759-1.00002-9.

Bom, Pedro R.D. and Jenny E. Ligthart (2014) "What Have We Learned from Three Decades of Research on the Productivity of Public Capital?" *Journal of Economic Surveys*, 28 (5), 889–916, 10.1111/joes.12037.

Caselli, Francesco and James Feyrer (2007) "The Marginal Product of Capital," *The Quarterly Journal of Economics*, 122 (2), 535–568.

de Mel, Suresh, David McKenzie, and Christopher Woodruff (2008) "Returns to Capital in Microenterprises: Evidence from a Field Experiment," *The Quarterly Journal of Economics*, 123 (4), 1329–1372.

Feenstra, Robert C., Robert Inklaar, and Marcel P. Timmer (2015) "The Next Generation of the Penn World Table," *American Economic Review*, 105 (10), 3150–3182, 10.1257/aer.20130954.

Jensen, Anders (2022) "Employment Structure and the Rise of the Modern Tax System," *American Economic Review*, 112 (1), 213–234, 10.1257/aer.20191528.

Kremer, Michael, Gautam Rao, and Frank Schilbach (2019) "Behavioral Development Economics," in *Handbook of Behavioral Economics: Applications and Foundations 1*, 2, 345–458: Elsevier, 10.1016/bs.hesbe.2018.12.002.

Lucas, Robert E. (1990) "Why Doesn't Capital Flow from Rich to Poor Countries?" *The American Economic Review*, 80 (2), 92–96.

Mauro, Paolo (1995) "Corruption and Growth," *The Quarterly Journal of Economics*, 110 (3), 681–712, 10.2307/2946696.

Solow, Robert M. (1956) "A Contribution to the Theory of Economic Growth," *The Quarterly Journal of Economics*, 70 (1), 65–94, 10.2307/1884513.